

Distributed Task Management Platform

Full-stack collaborative platform deployed on AWS EC2 at 99.9% uptime

Tech Stack | React.js · Node.js · Express.js · MongoDB · JWT · AWS EC2/S3 · GitHub Actions

Full Stack	AWS	CI/CD	RBAC	Real-Time Dashboard
------------	-----	-------	------	---------------------

Project Overview

- A production-deployed, full-stack collaborative task management platform built for teams. Hosted on AWS EC2, it sustains 100+ concurrent users at 99.9% uptime with JWT-based authentication, role-based access control (RBAC), and optimised MongoDB queries. Features real-time dashboards with priority queuing, deadline tracking, and per-user analytics.

Key Technical Highlights

- Deployed on AWS EC2 with sustained performance across 100+ concurrent users — validated through load testing.
- JWT authentication with RBAC ensures secure, role-scoped access: admins, managers, and contributors each see tailored views.
- MongoDB query optimisation via compound indexing on user-task relationships, reducing average query latency significantly.
- Real-time dashboards with priority queuing logic and deadline heatmaps surfacing at-risk tasks to team leads.
- Per-user analytics tracking task completion velocity and workload distribution across team members.
- Fully automated CI/CD pipeline on GitHub Actions — lint, test, build, and deploy triggered on every merge to main.

Architecture & Design Decisions

- React.js SPA frontend consuming Express.js REST APIs; API responses paginated and cached for dashboard performance.
- MongoDB schema designed with compound indexes on (userId, status, deadline) for fast filtered queries at scale.
- JWT tokens signed with RS256 asymmetric keys; RBAC middleware validates role claims on every protected route.
- AWS S3 used for static asset storage and file attachments linked to tasks; EC2 hosts the Node.js API server.
- GitHub Actions workflow: runs ESLint and Jest tests on PRs, builds Docker image on merge, deploys to EC2 via SSH.

- API contracts documented and deployment runbooks authored for zero-friction engineering handoff.

Outcomes & Impact

- Reduced team task completion lag by 40% through real-time priority dashboards and deadline tracking features.
- Zero downtime deployments achieved via automated CI/CD — every release validated by tests before hitting production.
- Scalable, maintainable codebase with full API documentation enabling rapid onboarding of new contributors.
- Demonstrated end-to-end ownership: system design, backend, frontend, cloud deployment, and DevOps pipeline.